3. We successfully implemented a real-time forecast system for up to five days, with good performance on Critical Success Index(CSI) over the forecast horizon.

2 Data

2.1 ERA5

For training our model, we built our datasets from ERA5 archive, the fifth generation of ECMWF reanalysis dataset (Hersbach et al., 2020). ERA5 is generated using ECMWF's Integrated Forecast System (IFS) cycle 42r1, operational for most of 2016. It employs an ensemble 4D-Var data assimilation scheme that incorporates 12-hour windows of observations from 21-09 UTC and 09-21 UTC, along with previous forecasts, allowing for a dense representation of the weather's state for each historical date and time. ERA5 assimilates high-quality global observations, making it widely regarded as the most accurate and comprehensive reanalysis archive.

We use a subset of ERA5 dataset on a 0.25° equiangular grid spanning 0-60° N, 70-140° E. The grid overall contains 241 × 281 grid points for latitude and longitude, respectively. 5 surface variables and 5 upper-air variables at 13 pressure levels (corresponding to the levels of the WeatherBench (Rasp & Thuerey, 2021) benchmark: 50, 100, 150, 200, 250, 300, 400, 500, 600, 700, 850, 925, and 1000 hPa). The 5 upper-air atmospheric variables are geopotential(Z), temperature(T), u component of wind(U), v component of wind(V), and specific humidity(SH). Additionally, 5 surface variables are 2-meter temperature(T2M), 10-meter u wind component(U10), 10-meter v wind component(V10), mean sea level pressure(MSLP), and total precipitation(TP).

2.2 CMPA

A high-resolution precipitation dataset, CMPA, was utilized in this study (Xie & Xiong, 2011). Developed by the China Meteorological Administration (CMA), CMPA combines multiple data sources, including observations from 30,000 to 40,000 ground-based stations, radar-derived quantitative precipitation estimates, and satellite precipitation retrievals from FY-2E and CMORPH (CPC MORPHing technique). By merging these sources, CMPA enhances the accuracy and resolution of precipitation measurements. It provides data at a high spatiotemporal resolution of 0.01° × 0.01° and 1-hour intervals, covering the region from 15°N to 60°N and 70°E to 140°E (4500 latitude × 7000 longitude grid points).

To reduce computational burden, we applied nn.AvgPool2d with PyTorch to transform the dataset into a lower resolution of 0.05° × 0.05°.

2.3 ECMWF real-time forecasts

Since February 2024, ECMWF has made open real-time medium-range forecast data (ECMWF, 2023) from the Integrated Forecasting System (IFS) available at an improved resolution of 0.25° × 0.25°. This dataset is publicly available and is well-suited for machine learning applications, providing a valuable resource for enhancing forecast accuracy. To generate high-resolution ensemble precipitation predictions, we use the forecast data from the 00 and 12 UTC initializations, spanning the next 120 hours at 3-hour intervals.

2.4 Preprocessing

All other variables are normalized using the mean and standard deviation except for precipitation,. Specifically, for precipitation, we first convert the 1-hour accumulated precipitation TP (mm) to reflectivity (dBZ) using the following formula (Marshall & Palmer,

1948):

$$dBZ = 10 \times \log_{10}(200 \times TP^{1.6}) \tag{1}$$

Afterward, the ERA5 and CMPA precipitation data are typically normalized to the [0, 1] range using the spatial maximum averaged over time, following the method outlined by Bodnar et al. (2024):

$$dBZ^{scale} = \frac{1}{N}\sum_{t=1}^{N} \max\{X_{t,i,j} : i = 1, \ldots, H; j = 1, \ldots, W\} \tag{2}$$

3 Methodology

3.1 Problem formulation

Inspired by the approach used in numerical weather models, where physical variables in the atmospheric governing equations can be represented as a combination of mean values and perturbations. The mean values correspond to the grid point values, and the perturbations account for sub-grid scale processes. These sub-grid processes are represented using physical parametrization schemes.

$$X_t = \bar{X}_t + X'_t \tag{3}$$

Similarly, we assume that high-resolution precipitation is composed of the average pre-

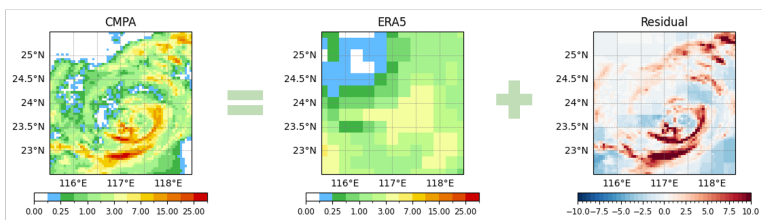

Figure 1. Decomposition of high-resolution precipitation

cipitation over a meso-scale $\overline{TP}$, plus the residual precipitation at the sub-grid scale TP'. We derive the residual precipitation TP' from the difference between scaled CMPA and ERA5 precipitation. ERA5 precipitation was interpolated to a 0.05° grid resolution using the nearest-neighbor method to match CMPA's resolution. As shown in Figure 2, we utilize a deterministic component for the mean precipitation and a latent diffusion component for the residual precipitation. Considering that processes on the sub-grid scale often exhibit randomness and uncertainty, we employ ensemble forecasting to capture and quantify these uncertainties. We run multiple stochastic sampling processes through the conditional diffusion model. Each sample represents a possible realization of the precipitation field, conditioned on the same input atmospheric states.

$$E = \{TP_i | \overline{TP} + TP'_i, i \in [1, 11]\} \tag{4}$$

3.2 Deterministic model

3.2.1 Model input

The deterministic model uses four ERA5 surface variables (T2m, U10m, V10m, and MSLP, with a shape of 4 × 241 × 281) and upper air variables at 13 pressure levels (T,

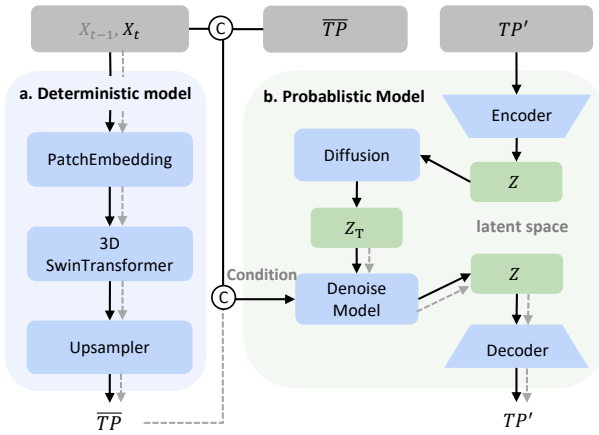

Figure 2. Overall model architeture. X_{t-1}/X_t includes surface variables (T2m, U10m, V10m, MSLP) and upper-air variables at 13 pressure levels (T, U, V, SP, Z) at time $t-1/t$. For the deterministic model, the inputs are X_{t-1} and X_t, while for the probabilistic model, the input is X_t. The model is trained on ERA5 data but can take forecasts from other models during inference. $\overline{TP}$ is the mean precipitation at time t, TP' is the residual precipitation. Z is the latent space, and Z_T is the latent variable at diffusion step T. The circle with "C" represents concatenation, black arrows represent the training process, and gray dashed lines indicate the inference process.

U, V, SP, and Z, with a shape of $5 \times 13 \times 241 \times 281$) of two consecutive time steps ($[X_{t-1}, X_t]$) to predict 1-hour accumulated precipitation at X_t. In addition to these variables, we also incorporated static features such as the binary land-sea mask, geopotential at the surface, soil type, latitude and longitude in radians, as well as temporal features including the sine and cosine of the local time of day and the sine and cosine of the yearly progress. These additional features were shown to improve the diagnostic prediction of precipitation, as demonstrated in the 4.1.

3.2.2 Model architecture

The deterministic model architecture consists of three main components:

1. Patch embedding, which divides the input variables into small patches and embeds them into high-dimensional feature representations.
2. 3D Swin-Transformer, which processes the embedded data by capturing both local and global features through multi-head self-attention mechanisms in spatial and temporal dimensions.
3. Upsampler, which process the output of the 3D-Swin Transformer to produce the final precipitation prediction at the original data resolution.

In a standard patch embedding for vision transformers, convolution is typically used to project the input data into embedded patches. However, this approach produces outputs that reflect only the linear relationships between the variables after convolution. To address this, we propose an non-linear approach: applying separate convolutions to each variable individually, followed by a GELU (Gaussian Error Linear Unit) activation. The outputs are then passed through an MLP layer and another GELU activation, ensuring that each patch integrates nonlinear information from the variables. We use a 4×4 convolution kernel for surface variables and a $2 \times 4 \times 4$ convolution kernel for upper air variables. This results in an output with a shape of $C\times1\times61\times71$ for the surface vari-